

KRISHIDRISHTI AI

Full-Stack Satellite & ML Monitoring Platform for Agriculture & Water Resources

Product Requirements Document

Problem Statement 85 | Version 4.0 — Expanded End-to-End Build

Field	Detail
Document Owner	Product Management — KrishiDrishti AI
Status	Draft for Engineering & Stakeholder Review
Version	4.0 (supersedes v3.0 Final End-to-End Build)
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Primary Domains	Remote Sensing, Geospatial Analytics, Applied ML, AgriTech

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1. Executive Summary & Vision

1.1 The Core Problem

Large-scale monitoring of crop conditions and water resources today relies on manual field surveys — slow, inconsistent, and resource-intensive. Surveys are typically completed weeks after a stress event begins, and coverage is sparse (a sampled percentage of plots, not all of them). By the time a drought, pest outbreak, or reservoir depletion is officially reported, yield loss or water scarcity has already compounded, leaving governments reactive and farmers under-informed.

1.2 The Solution

KrishiDrishti AI is a unified, end-to-end Geospatial Analytics Platform that ingests raw multi-spectral satellite imagery on a recurring schedule, derives remote-sensing indices (NDVI for vegetation health, NDWI for water bodies), visualizes results on an interactive map, and layers Machine Learning on top to forecast crop yield and flag water/drought stress before it becomes a crisis. The platform shifts agricultural and water-resource management from reactive reporting to predictive, plot-level decision-making — for three distinct audiences (government, farmers, and insurers) from one shared data backbone.

1.3 Why Now

- Sentinel-2 and Landsat imagery is freely available at 10m resolution with a 5-day revisit cycle, making near-real-time monitoring economically viable for the first time.
- Cloud-based geospatial processing (Google Earth Engine, Sentinel Hub) removes the need to store and process raw satellite tiles in-house.
- Smartphone and SMS penetration in rural India makes direct farmer alerts operationally feasible.
- Crop insurance schemes (e.g. PMFBY) increasingly require objective, third-party yield-loss evidence — a natural commercial anchor for this platform.

1.4 Product Principles

- **Predictive, not reactive:** every insight should exist before the damage is visible on the ground, not after.
- **One dataset, three audiences:** the same pipeline serves government dashboards, farmer alerts, and insurer evidence — no duplicated ingestion.
- **Explainable AI:** every ML prediction is traceable to the NDVI/NDWI/weather inputs that produced it — critical for insurance disputes and farmer trust.
- **Low-bandwidth first:** core farmer-facing flows (alerts, plot status) must work over SMS and 2G, not just a rich web dashboard.

2. Goals & Success Metrics

2.1 Business Goals

- Reduce the average time-to-detection of crop/water stress from weeks (manual survey) to under 5 days (satellite revisit cycle).

- Provide government nodal officers with district-wide drought/water-depletion visibility without a single field visit.
- Give insurers an objective, auditable data trail to accelerate and de-risk claim settlement.
- Establish a reusable geospatial + ML platform that can extend to new indices (soil moisture, pest risk) without re-architecture.

2.2 Key Performance Indicators (KPIs)

KPI	Target (Phase 1)	Target (Phase 2)
AOI (plot/lake) processing latency	< 8 sec for a 100-ha polygon	< 5 sec at 5x concurrent load
Concurrent polygon processing requests	50 simultaneous	200 simultaneous
Yield prediction accuracy (MAPE)	< 20% mean absolute % error	< 12% mean absolute % error
Early Warning alert delivery time	< 15 min from model inference	< 5 min
Dashboard first-load time	< 2 sec	< 1.5 sec
Registered farmer plots (adoption)	5,000 plots in pilot district	50,000 plots across state
SMS alert delivery success rate	> 95%	> 98%
Platform uptime	99.5%	99.9%

3. Target User Personas & Jobs-To-Be-Done

Each persona below expands the original one-line need into a job-to-be-done, concrete pain points, and the specific platform capabilities that address them — used to trace every feature in Section 5 back to a real user.

3.1 Government / Nodal Officer — “Anita, District Agriculture Officer”

MACRO VIEW

Job to be done: “When water levels or crop health decline across my district, I want to know within days — not after farmers report distress — so I can trigger relief measures and justify drought declarations with data.”

Pain points today: Field surveys cover a sample of villages; drought declarations arrive 4-6 weeks after onset; no single source of truth across taluks.

Platform solution: District-wise aggregating dashboards, automated lake surface-area tracking, exportable evidence reports for official declarations.

3.2 Farmer / FPO Member — “Ramesh, 5-acre cotton farmer”

MICRO VIEW

Job to be done: “When my crop starts to stress, I want to be told in my language, on my phone, in time to irrigate or treat it — before I lose the season.”

Pain points today: No visibility into crop health until visible wilting; irrigation decisions made by habit, not data; no low-bandwidth way to check plot status.

Platform solution: Mobile-responsive UI, custom polygon drawing for individual plots, SMS alerts in regional language for early stress detection.

3.3 Insurance Company — “Claims team, agricultural insurer”

VERIFICATION

Job to be done: “When a farmer files a crop-loss claim, I want objective, time-stamped, third-party evidence so I can settle fairly and quickly without a manual field inspection.”

Pain points today: Field inspections are slow and subject to dispute; no historical baseline to compare against; fraud risk from self-reported loss.

Platform solution: Historical satellite time-series per plot, ML yield-prediction logs with model version and input snapshot, exportable audit-ready reports.

3.4 Platform Administrator (new persona)

OPERATIONS

Job to be done: “I need to manage user accounts, monitor pipeline health, and retrain/redeploy ML models without touching production code.”

Platform solution: Admin console for user/role management, pipeline monitoring dashboard (Module 6), model version registry.

4. Scope

4.1 In Scope (Phase 1 — MVP)

- AOI drawing + administrative boundary search, NDVI/NDWI raster generation, temporal slider, yield prediction for a single crop type (cotton, as pilot), Early Warning alerts via SMS, PDF report export, role-based access for the three core personas.

4.2 In Scope (Phase 2+)

- Multi-crop yield models, pest/disease risk layer, soil moisture index, mobile app (native), insurer API integration for automated claims triggers, multi-state rollout.

4.3 Out of Scope (explicitly)

- Drone/UAV imagery ingestion (satellite-only for Phase 1-2).
- Direct payment/subsidy disbursement — the platform provides evidence, not the payment rail.
- On-device offline map caching (dependent on connectivity for Phase 1).

5. Functional Requirements

Each requirement carries a unique ID (Module.Sequence) for traceability into engineering tickets and QA test plans.

Module 1 — Interactive GIS Dashboard (Frontend)

The visual interface where all three personas interact with the data. Built as a map-first, panel-second layout.

F1.1 Area of Interest (AOI) Toolkit

Users can manually draw a polygon (custom farm or lake boundary) or search/select standard administrative boundaries (Village → Taluk → District) from a hierarchical picker.

Acceptance Criteria:

- Polygon drawing supports at least 500 vertices without UI lag.
- Drawn polygons persist to the user's account and are re-selectable from a “My AOIs” list.
- Administrative boundary search returns results in under 500ms for a partial-name query.
- Area of a drawn polygon is displayed live in hectares/acres while drawing.

F1.2 Multi-Layer Map Engine

Base maps (Street, Topographic, Satellite) with toggleable raster overlays: True Color (RGB), NDVI (Vegetation), and NDWI (Water). Layers can be blended with adjustable opacity.

Acceptance Criteria:

- Layer toggle re-renders the visible AOI in under 3 seconds.
- Opacity slider ranges 0-100% with live preview.
- A legend is always visible for any active index layer (color → value mapping).

F1.3 Temporal Slider (Time-Machine)

A UI slider allowing users to smoothly scrub through imagery from different dates (e.g., watching a lake shrink from January to May), snapping to actual satellite pass dates.

Acceptance Criteria:

- Slider only exposes dates where cloud-free imagery exists for the selected AOI (no dead frames).
- Scrubbing between two adjacent dates completes the layer swap in under 1 second (pre-fetched/cached tiles).
- A “compare” mode supports side-by-side (split-screen) view of two selected dates.

F1.4 District/State Aggregate Dashboard

Government officers see a roll-up view: number of plots by stress category, total water-body surface area vs. baseline, and drought-risk-flagged polygons, aggregated at Taluk and District level.

Acceptance Criteria:

- Aggregate counts recompute automatically as new satellite passes are ingested (no manual refresh required).
- Drill-down from District → Taluk → Village → individual plot is supported in 3 clicks or fewer.

F1.5 Localization (i18n) Switcher

UI language toggle supporting English, Hindi, and at least one regional language at launch (configurable per pilot state).

Acceptance Criteria:

- All farmer-facing screens and SMS templates are translated, not just labels (units, date formats localized too).
- Language preference persists per user account.

Module 2 — Satellite Remote Sensing Engine (Backend Processing)

The core calculation engine that turns raw multi-spectral images into actionable indices.

F2.1 Automated Data Pipeline

A scheduled ingestion job hooks into Google Earth Engine / Sentinel Hub to pull the latest cloud-free optical and infrared imagery for all active AOIs.

Acceptance Criteria:

- Pipeline runs on a configurable schedule (default: every 5 days, matching Sentinel-2 revisit).
- Cloud-cover filtering discards scenes above a configurable threshold (default 20%) and logs the gap rather than silently failing.
- Failed ingestion jobs auto-retry 3 times with exponential backoff before alerting the on-call admin.

F2.2 Vegetation Stress (NDVI) Engine

Computes NDVI per pixel within an AOI and classifies fields into Green (High Health, NDVI > 0.6), Yellow (Moderate Stress, 0.3-0.6), or Red (Severe Stress, < 0.3), with thresholds configurable per crop type.

Acceptance Criteria:

- NDVI computation for a 100-hectare AOI completes in under 5-8 seconds (per NFR).
- Classification thresholds are stored per-crop in a config table, not hardcoded.
- Each computed NDVI raster is stored with its source scene ID and acquisition date for auditability.

F2.3 Water Depletion (NDWI) Engine

Dynamically segments water boundaries and calculates the exact surface area of lakes/reservoirs in hectares/acres, compared against a rolling baseline (e.g., 5-year average for the same calendar week).

Acceptance Criteria:

- Surface-area calculation error is within 5% of ground-truth validation samples.
- A depletion percentage vs. baseline is computed and stored per water body per satellite pass.
- A configurable depletion threshold triggers a flag for Module 3's Early Warning System.

F2.4 Cloud/Shadow Masking

Pre-processing step that masks cloud, cloud-shadow, and haze pixels before index calculation to prevent false stress readings.

Acceptance Criteria:

- Masked pixels are excluded from NDVI/NDWI aggregate statistics, not counted as zero/stressed.
- If more than 40% of an AOI is masked in a given pass, that pass is flagged “insufficient coverage” rather than shown as data.

Module 3 — Machine Learning & Predictive AI

The forecasting layer that adds intelligence to raw remote-sensing data.

F3.1 Crop Yield Prediction Model

A Random Forest Regressor trained on historical NDVI, NDWI, and localized weather data (rainfall, temperature) predicts end-of-season yield (kg/hectare) per AOI, per crop.

Acceptance Criteria:

- Model exposes a confidence interval alongside the point prediction, not a bare number.
- Model version and feature snapshot used for each prediction are logged for reproducibility (needed for insurer audit trails).

- Retraining pipeline can be triggered on a schedule or manually from the Admin console (Module 6).

F3.2 Early Warning System (EWS)

If the ML model predicts a yield drop of more than 20% vs. the 5-year average for that AOI, the system automatically raises a “Drought Risk Alert” and queues Module 4 notifications.

Acceptance Criteria:

- Alert threshold (20%) is configurable per crop/region, not hardcoded globally.
- Each alert records the triggering prediction, input snapshot, and timestamp for later review.
- Duplicate alerts for the same AOI are suppressed within a 5-day cooldown window.

F3.3 Model Explainability Panel

For any yield prediction, the UI can show the top contributing factors (e.g., “low NDVI in weeks 6-8”, “rainfall 30% below average”) using feature-importance output from the model.

Acceptance Criteria:

- Explainability view loads for any historical prediction, not just the most recent.
- Output is phrased in farmer-readable language on the mobile view, and technical (SHAP/feature-importance values) on the admin/insurer view.

Module 4 — Reporting, Alerts & Notifications

F4.1 Automated PDF Reports

One-click generation of a summary report for any selected AOI, containing NDVI/NDWI charts, current index scores, ML predictions, and (for insurers) the full time-series evidence trail.

Acceptance Criteria:

- Report generation completes in under 10 seconds for a single AOI.
- Report includes a unique report ID and generation timestamp for audit reference.
- Report template varies by persona: Farmer (simple, localized), Government (district roll-up), Insurer (full audit trail).

F4.2 SMS/Email Notifications

Integration with Twilio/AWS SNS to alert registered farmers when their farm enters the “Severe Stress” zone or an Early Warning is triggered, in their selected language.

Acceptance Criteria:

- SMS delivery success is tracked and retried once on failure; failure after retry is logged and surfaced in the Admin console.
- Notification includes a short actionable recommendation (e.g., “Irrigate within 48 hours”), not just a raw alert.
- Farmers can opt in/out of SMS vs. email vs. both per account.

F4.3 In-App Notification Center

A persistent notification feed inside the web/mobile UI mirroring SMS/email alerts, with read/unread state and historical alert log per AOI.

Acceptance Criteria:

- Notification center shows the last 90 days of alerts per AOI.
- Clicking a notification deep-links to the relevant AOI and date on the map.

Module 5 — User, Access & Account Management (new)

F5.1 Authentication & Role-Based Access Control

Secure login (email/phone + OTP or password) with three primary roles — Farmer, Government Officer, Insurer — plus an Admin role, each with scoped data visibility.

Acceptance Criteria:

- Farmers can only view/edit their own AOIs; Government Officers have read-only district-scoped access; Insurers have read-only access limited to AOIs linked to an active policy.
- Passwords are never stored in plaintext; OTP flows expire in 5 minutes.
- Session tokens expire after 24 hours of inactivity.

F5.2 FPO / Organization Grouping

Farmers can be grouped under a Farmer Producer Organization (FPO), letting an FPO coordinator view aggregate stress status across member plots.

Acceptance Criteria:

- FPO coordinator view aggregates without exposing individual farmer contact details unless explicitly shared.

Module 6 — Admin & Platform Operations Console (new)

F6.1 Pipeline Health Monitoring

A dashboard for admins showing ingestion job status, queue depth (Celery/Redis), failed-task counts, and per-region data freshness (last successful pass per AOI cluster).

Acceptance Criteria:

- Any AOI cluster with no successful ingestion in over 10 days is flagged in red.
- Admin can manually re-queue a failed ingestion job from the console.

F6.2 ML Model Registry

A version-controlled registry of trained `.pkl` models with metadata (training date, dataset version, validation MAPE) and one-click rollback to a prior model version.

Acceptance Criteria:

- Every production inference records which model version served it.
- Rollback to a previous model version takes effect within 5 minutes without a full deployment.

6. Full-Stack System Architecture

To support high-computation satellite processing and ML inference without blocking the UI, the architecture is distributed across six layers, expanded below with the responsibility and key interfaces of each.

Layer	Primary Technologies	Responsibility
1. External Data Sources	Google Earth Engine (GEE), Sentinel Hub, OpenWeather API	Provides raw multi-spectral imagery and localized climate data on a scheduled pull.
2. Core Backend API	Python (FastAPI), GeoPandas, Rasterio	Handles API requests, authentication, spatial calculations, and orchestrates async jobs.
3. Async Task Queue	Celery + Redis	Runs heavy raster processing and ML inference in the background so the UI never blocks; supports 50+ concurrent polygon jobs.
4. AI/ML Engine	Scikit-Learn, Pandas, Joblib, MLflow (model registry)	Serves pre-trained `.pkl` predictive models and logs prediction provenance.
5. Database Layer	PostgreSQL + PostGIS extension, Redis (cache)	Stores user profiles, drawn polygons, time-series index history, alerts, and report metadata.
6. Client Frontend	React.js / Next.js, Mapbox GL JS, Tailwind, Chart.js	Renders interactive maps, charts, and role-specific dashboards; mobile-responsive.

6.1 Request Flow (High Level)

1. Client draws/selects an AOI → request hits the FastAPI backend.
2. Backend validates the AOI (geometry, size limits) and enqueues a Celery task.
3. Celery worker fetches the latest cloud-free scene from GEE/Sentinel Hub, runs cloud masking, and computes NDVI/NDWI via Rasterio/GeoPandas.
4. Computed rasters + statistics are written to PostgreSQL/PostGIS; tile outputs are cached (Redis / object storage) for fast map rendering.
5. The ML Engine pulls the latest index values + weather data, runs the Random Forest model, and stores the prediction with model version metadata.
6. If the prediction crosses the Early Warning threshold, an alert event is published, consumed by the Notification service (Twilio/SNS).
7. Client polls or receives a websocket push when the job completes, then renders the updated layer and side panel.

6.2 Why This Architecture

- Separating the task queue (Layer 3) from the API (Layer 2) means a spike in raster-processing demand never degrades login, search, or dashboard responsiveness.
- PostGIS was chosen over a generic document store because polygon intersection, area calculation, and spatial indexing are native, indexed operations — critical at 50+ concurrent AOI queries.
- A model registry (MLflow or equivalent) rather than a single deployed `.pkl` file is required so predictions remain reproducible for insurance audits even after the model is retrained.

7. Data Model (Core Entities)

Simplified entity overview; full schema (columns, indexes, foreign keys) to be maintained in a separate technical design doc, but the core relationships below drive every module above.

Entity	Key Attributes	Notes
User	id, role (farmer/gov/insurer/admin), phone, email, language_pref, org_id	org_id links Farmers to an FPO (nullable).
AOI (Polygon)	id, owner_id, geometry (PostGIS), type (farm/lake/admin_boundary), crop_type, created_at	geometry indexed with GiST for fast spatial queries.
SatellitePass	id, aoi_id, scene_id, acquisition_date, cloud_cover_pct, source (Sentinel-2/Landsat)	One row per successful ingestion per AOI.
IndexResult	id, pass_id, index_type (NDVI/NDWI), mean_value, classification, raster_uri	raster_uri points to cached tile storage, not inline blob.
YieldPrediction	id, aoi_id, model_version, predicted_yield_kg_ha, confidence_interval, input_snapshot_json, created_at	input_snapshot_json preserves exact features used — audit requirement.
Alert	id, aoi_id, type (drought_risk/severe_stress), triggering_prediction_id, status, created_at	status: open / notified / acknowledged.
Notification	id, alert_id, channel (sms/email), delivery_status, sent_at	Retried once on failure; logged either way.
Report	id, aoi_id, requested_by, persona_template, file_uri, generated_at	file_uri points to generated PDF in object storage.

8. Representative API Endpoints

Illustrative REST surface (FastAPI) — not exhaustive, but enough to validate the module boundaries above map cleanly onto a service contract.

Method & Path	Purpose	Primary Consumer
POST /aois	Create a new AOI (drawn polygon or admin boundary reference)	Web/Mobile client
GET /aois/{id}/timeline	List available satellite pass dates for the Temporal Slider	Web client
GET /aois/{id}/index?type=NDVI&date=	Fetch a specific index raster/stats for one date	Map layer renderer
POST /aois/{id}/predict	Trigger (or fetch cached) yield prediction for an AOI	Web/Mobile client

Method & Path	Purpose	Primary Consumer
GET /aois/{id}/predict/history	Retrieve prediction history with model versions	Insurer audit view
GET /districts/{id}/summary	Aggregate stress/water counts for a district	Government dashboard
POST /reports	Queue a PDF report generation job	Web/Mobile client
GET /alerts?aoi_id=	List alert history for an AOI	Notification center
GET /admin/pipeline/status	Pipeline health & queue depth	Admin console
POST /admin/models/{id}/rollback	Roll back to a prior model version	Admin console

9. Non-Functional Requirements (NFRs)

9.1 Scalability & Concurrency

- The system must handle at least 50 simultaneous polygon-processing requests at launch, scaling to 200 by Phase 2, achieved via horizontally-scaled Celery workers.
- Database connection pooling and read replicas must be in place before the district aggregate dashboard (F1.4) is enabled for a full state rollout.

9.2 Performance & Latency

- Dashboard UI must load within 2 seconds on a 4G connection.
- Raster layer rendering over a 100-hectare farm must complete within 5-8 seconds end-to-end (request to rendered tile).
- Temporal Slider frame swaps between cached dates must complete in under 1 second.

9.3 Data Security & Privacy

- All user data (farm locations, contact details) must be encrypted in transit (HTTPS/TLS 1.2+) and at rest (AES-256).
- Role-based access control (F5.1) enforced at the API layer, not just hidden in the UI.
- Farmer PII (phone, precise plot geometry) is not exposed to Insurer or Government roles beyond what is contractually required.

9.4 Availability & Reliability

- Target uptime: 99.5% in Phase 1, 99.9% in Phase 2, excluding scheduled maintenance windows.
- Failed satellite ingestion must degrade gracefully — the UI shows “last known data from [date]” rather than an error state.

9.5 Localization (i18n)

- UI, PDF reports, and SMS templates must support English, Hindi, and regional languages configurable per deployment state.

9.6 Accessibility

- Web dashboard meets WCAG 2.1 AA for color contrast — notably important given NDVI's red/yellow/green palette must remain distinguishable for color-blind users (a colorblind-safe alternate palette should be selectable).

9.7 Auditability

- Every yield prediction and alert must be traceable to the exact model version and input data snapshot used — a hard requirement for insurance claim disputes.

10. End-to-End User Journeys

10.1 Farmer Journey — Early Warning to Action

8. Ramesh logs into the mobile-responsive web portal and draws a boundary around his 5-acre cotton farm (F1.1).
9. The backend queues a Celery task; it fetches the latest cloud-free Sentinel-2 imagery and computes NDVI (F2.2).
10. The backend also pulls local weather data and feeds both into the pre-trained Random Forest yield model (F3.1).
11. The map updates with a red/yellow/green heatmap over the farm. A side panel shows: “Current Health: Moderate Stress. AI Prediction: 15% yield loss expected due to recent low moisture. Recommendation: Irrigate within 48 hours.” (F3.3)
12. Because the predicted loss is under the 20% Early Warning threshold, no SMS fires yet — but the in-app notification center logs it (F4.3).
13. Ramesh downloads a simplified, Hindi-language PDF report to share with his agronomist (F4.1).

10.2 Government Officer Journey — District Drought Declaration

14. Anita opens the District Aggregate Dashboard and sees 34% of monitored plots in her district flagged Yellow/Red this week (F1.4).
15. She drills down Taluk → Village and identifies two Taluks with reservoir surface area 40% below the 5-year baseline (F2.3).
16. She exports a district-level evidence report (F4.1, Government template) to support an official drought declaration request.

10.3 Insurer Journey — Claim Verification

17. A farmer files a crop-loss claim referencing their registered AOI.
18. The insurer's claims team pulls the AOI's full historical satellite time-series and yield-prediction log, including model version and input snapshot for the claim period (F3.1, F8 API).
19. The claims team cross-references the Early Warning alert history (F3.2) to confirm the loss pattern is consistent with satellite-observed stress, and settles the claim without a field visit.

11. Release Roadmap

Phase	Duration	Scope
Phase 0 — Foundations	Weeks 1-2	Architecture setup (Layers 1-6 skeleton), GEE/Sentinel Hub integration, PostGIS schema, auth (F5.1).
Phase 1 — MVP	Weeks 3-8	Modules 1 & 2 full build, single-crop yield model (F3.1), EWS (F3.2), SMS alerts (F4.2), PDF reports (F4.1), pilot district launch.
Phase 2 — Scale & Intelligence	Weeks 9-16	Multi-crop models, model explainability (F3.3), admin console (Module 6), 200-concurrency scaling, multi-state rollout.
Phase 3 — Ecosystem	Weeks 17+	Insurer API integration, native mobile app, soil moisture/pest-risk layers, FPO grouping (F5.2).

12. Risks & Mitigations

Risk	Impact	Mitigation
Persistent cloud cover blocks optical imagery in monsoon season	No data for weeks during the highest-risk period	Fall back to SAR (Sentinel-1) imagery for water/flood detection when optical coverage fails; surface “stale data” state honestly rather than guessing.
Yield model accuracy degrades outside training region/crop	Wrong Early Warning alerts erode farmer trust	Launch with a single well-validated crop/region (cotton, pilot district) before generalizing; always show confidence interval.
SMS delivery failure in low-network rural areas	Farmers miss time-critical alerts	Dual-channel (SMS + in-app) with delivery-status tracking and retry; explore IVR fallback in Phase 2.
Insurer disputes a claim decision based on model output	Legal/trust exposure	Full audit trail (model version + input snapshot) stored immutably per prediction; explainability panel (F3.3).
Concurrent load exceeds 50 polygon jobs at pilot scale-up	Processing delays, missed SLAs	Load-test Celery worker autoscaling before district-wide launch; queue-depth alerting in admin console.

13. Assumptions & Open Questions

13.1 Assumptions

- Google Earth Engine / Sentinel Hub access and quota are secured before Phase 0 begins.

- Pilot district and pilot crop (cotton) are confirmed by the government partner before Phase 1 kickoff.
- Twilio/AWS SNS account and sender-ID registration (required for India SMS compliance) are in place before Phase 1 SMS testing.

13.2 Open Questions for Stakeholder Review

20. What is the legal/contractual basis for insurers to access farmer plot data — opt-in per policy, or blanket agreement with the FPO?
21. Should the 20% Early Warning threshold be uniform, or set per state agriculture department policy?
22. Is offline/low-connectivity support (cached last-known state) required for Phase 1, or acceptable to defer to Phase 3?
23. Who owns model retraining cadence and validation sign-off — the ML team, or a joint review with the agriculture domain expert?